

Quantum Mechanics in Solids

Unit-5

Lecture-2

Free electron theory of metals

Properties of Metals Explained by this theory: -

① Ohm's Law

$$V \propto I \longrightarrow \vec{J} = \sigma \vec{E}$$

② For pure Metals

$$\sigma \propto \frac{1}{T}$$

③ Wiedmann-Franz Law

Diagram illustrating the Wiedmann-Franz Law:

$$\frac{K}{\sigma} \propto T$$

Labels and arrows:

- $\frac{K}{\sigma}$ is labeled as "Coeffⁿ of Thermal Conductivity".
- σ is labeled as "Electrical Conductivity".
- T is labeled as "Absolute tempⁿ".

Electrical Conductivity (σ) :-

$$J = \sigma E$$

$$\frac{I}{A} = \sigma E$$

$$\sigma = \frac{I}{EA} = \frac{Q}{tEA}$$

accelⁿ of free Electron

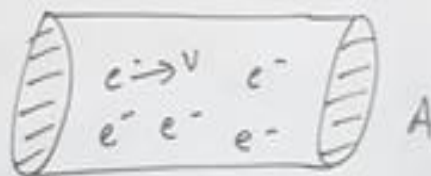
$$a = \frac{d^2x}{dt^2} = \frac{eE}{m}$$

Integrate

$$\frac{dx}{dt} = V = \frac{eE}{m}t + C$$

At $t=0$, $V=0$, $C=0$

$$V = \frac{eEt}{m} \quad (2)$$



Energy of free Electron

$$KE = \frac{1}{2}mv^2$$

Accⁿ to kinetic Theory

$$KE = \frac{3}{2}k_B T$$

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

$$mv = \frac{3k_B T}{V} \quad (3)$$

Relaxation Time

$$\tau = \frac{\lambda}{V} \leftarrow \text{Mean Free Path}$$

Average Velocity

$$\bar{V} = \frac{\int_0^\tau V dt}{\int_0^\tau d\tau} = \frac{1}{\tau} \int_0^\tau \frac{eEt}{m} dt$$

$$\bar{V} = \frac{eE}{m\tau} \int_0^\tau t dt = \frac{eE}{m\tau} \left[\frac{t^2}{2} \right]_0^\tau$$

$$\bar{V} = \frac{eE\tau}{2m} = \frac{eE\lambda}{2mV} = \frac{eE\lambda}{2 \times 3k_B T \frac{\lambda}{V}}$$

$$\bar{V} = \frac{eE\lambda V}{6k_B T} \quad (4)$$

We know that

$$J = n e \bar{v}$$

↑
Electron Density

$$J = n e \times \frac{e E \lambda v}{6 k_B T}$$

$$J = \left(\frac{n e^2 \lambda v}{6 k_B T} \right) E$$

$$\boxed{J \propto E}$$

$$\boxed{J = \sigma E}$$

→ Vector form
of ohm's law

$$\sigma = \frac{n e^2 v \lambda}{6 k_B T}$$

$$\boxed{\sigma \propto \frac{1}{T}}$$

Thermal Conductivity (K):-

$$Q \propto A \left(\frac{dT}{dx} \right) t$$

Tempⁿ
Gradient



$$Q = -KA \frac{dT}{dx} t$$

coeffⁿ of
thermal conductivity

$$Q = K \frac{dT}{dx} \quad \text{--- (1)}$$

Now, Accⁿ to Kinetic Theory of Gases,
Amount of heat passes through b/w
two unit area surfaces for unit time

$$Q = \frac{1}{3} n v \lambda \frac{dE}{dx} \quad \text{--- (2)}$$

n = Electron Density

v = velocity of E^-

$\frac{dE}{dx}$ = Energy Gradient

$$E = \frac{3}{2} k_B T$$

$$\frac{dE}{dx} = \frac{3}{2} k_B \frac{dT}{dx}$$

$$Q = \frac{1}{3} n v \lambda \times \frac{3}{2} k_B \frac{dT}{dx} = \frac{1}{2} n v \lambda k_B \frac{dT}{dx} \quad \text{--- (3)}$$

Compare Eqⁿ (1) & (3)

$$K \frac{dT}{dx} = \frac{1}{2} n v \lambda k_B \frac{dT}{dx}$$

$$K = \frac{1}{2} n v \lambda k_B$$

Thermal Conductivity (K):-

$$\sigma = \frac{ne^2 v \lambda}{6 k_B T}$$

$$K = \frac{1}{2} n v \lambda K_B$$

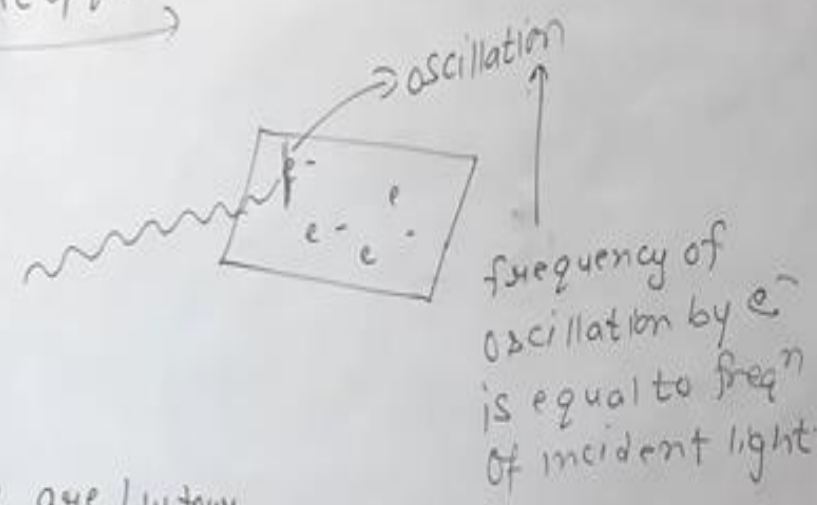
$$\frac{K}{\sigma} = \frac{\frac{1}{2} n v \lambda K_B}{\frac{ne^2 v \lambda}{6 k_B T}}$$

$$\frac{K}{\sigma} = \frac{1}{2} n v \lambda K_B \times \frac{6 k_B T}{ne^2 v \lambda}$$

$$\frac{K}{\sigma} = \left(\frac{3 k_B^2}{e^2} \right) T$$

$\left[\frac{K}{\sigma} \propto T \right] \rightarrow$ Verify Wiedmann Franz Law

Metals are opaque



Metals are Lustrous



free electrons gain Energy & move to higher Energy level

